UNIT 10 · TOPIC 2

Geographical Concepts and Field Variability

Syllabus topic: Geographical concepts of precision agriculture; understanding and identifying variability

Identifying variability

PA begins by identifying how a field varies:

Variability	Examples	Captured by
SOIL	Texture, nutrients, pH, moisture, organic matter, depth	Georeferenced sampling, soil EC sensing
YIELD	Grain yield across the field	Yield monitors (Topic 5)
TOPOGRAPHIC	Elevation, slope	Elevation surveys, GPS
CROP / canopy	Vigour, disease, weeds	Sensors and imagery (Topic 6)
MANAGEMENT	Past practices, drainage, traffic	Records

Variability is captured by georeferenced sampling and sensing, then displayed as MAPS (Topic 8).

Geographical concepts

- Every observation is **GEO-REFERENCED** — latitude/longitude or projected coordinates.
- Spatial data are **RASTER** (grid cells — e.g., imagery) or **VECTOR** (points, lines, polygons).
- Accurate positioning (GPS — Topic 3) is what makes site-specific mapping and application possible.

Practice MCQs

Q7. Spatial data as grid cells (e.g., imagery) are:

(a) Vector (b) Raster
(c) Tabular (d) Analog

Q8. Points, lines and polygons are ___ data:

(a) Raster (b) Vector
(c) Pixel (d) Scalar

Answers

Q7 — (b) Raster

Q8 — (b) Vector

Sources: Srinivasan — Handbook of Precision Agriculture; NIIT/ICAR precision-agriculture modules; NovAtel — Introduction to GNSS; ISRO/PIB NavIC data (as given in the source notes, July 2026; operational status updated October 2026); NPTEL precision-agriculture lectures.